



Bring Your Own Cloud (BYOC)

Security Shared Responsibility Model

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ClickHouse Bring Your Own Cloud (BYOC)

Security Shared Responsibility Model

ClickHouse enables you to deploy services to your own cloud resources to improve control over your data. This deployment method relies on shared responsibilities between you (the customer), ClickHouse, and the cloud service providers. This document provides a high level breakdown of these responsibilities. For more information on specific features and settings you can use to meet your security needs, please visit our [Security Shared Responsibility Model](#) page or <https://trust.clickhouse.com>.

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ClickHouse Responsibilities

Customer Responsibilities

Cloud Responsibilities



IAM

- Enforce unique usernames, strong passwords, and multi-factor authentication (MFA)
- Restrict access to customer environments based on principle of least privilege
- Secure remote connections using strong cryptography
- Manage Identity and Access Management (IAM) holistically, including oversight of accounts BYOC customers create on Auth0

- Configure single sign-on (SSO) for console users and enforce multi-factor authentication within the identity provider
- Use strong passwords and configure roles based on least privilege for database users
- Securely manage the default user password and relevant API keys and secrets

- Protect the identity and access management infrastructure



Data Security

- Encrypt data in transit using TLS 1.2+
- Encrypt data at rest using AES 256+
- Securely manage, deploy, and rotate encryption keys
- Delete service data and backups within seven (7) days of service termination

- Implement customer-managed encryption keys (CMEK), as available
- Utilize time-to-live settings to enforce data retention

- Manage encryption hardware and services
- Encrypt data in transit and at rest, where configured



Network

- Deploy security groups and network controls to enable secure communication while isolating customer environments
- Enable secure defaults for network access controls and security groups

- Configure IP filters to restrict connections to the database
- Maintain secure network configurations after initial deployment

- Manage physical and logical security of the cloud networking infrastructure
- Maintain secure communications for cloud infrastructure, including APIs

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Security Monitoring

Manage monitoring and incident response for ClickHouse-managed environments:

- Deploy security event detection capabilities
- Generate audit logs and retain for 1 year
- Investigate and respond to potential security events
- Report security breaches affecting the customer in accordance with the ClickHouse Information Security Addendum

Manage monitoring and incident response for customer-managed environments:

- Configure and manage cloud security monitoring
- Monitor session and query logs within the service
- Investigate and respond to potential security events

- Configure and manage security monitoring for underlying cloud services
- Investigate and respond to potential security events related to underlying cloud services
- Report security breaches affecting the customer in accordance with contractual obligations



Disaster Recovery

- Protect against database failures by using multiple replicas
- Utilize multi-availability zone configurations in each region
- Provide backup capabilities to enable data recovery from localized incidents
- Regularly test backups to ensure recoverability

- Configure backup policies and perform restoration

- Provide data centers with high-availability features
- Provide geographically isolated data centers in each region

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Platform

- Securely configure, deploy, and terminate ClickHouse systems
- Utilize hardened base images to deploy services
- Maintain a public bug bounty program

- Secure the service landing zone, including account setup, configuration, and management

- Provide and maintain physical and environmental protections
- Securely configure, patch, and maintain hardware, firmware, and operating system software



Best Practices

- Maintain a technical vulnerability management program
- Conduct third-party penetration tests at least annually
- Employ an in-house information security team

- Configure ClickHouse and cloud security controls based on organizational requirements
- Follow security best practices for cloud-based systems

- Maintain a technical vulnerability management program
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Compliance

- Maintain independent third-party audits, standards, and certifications
- Provide tools and configurations that enable compliance with applicable laws, such as GDPR and CCPA

- Evaluate and implement relevant ClickHouse security configurations to meet applicable compliance requirements for the type of data processed
- Use ClickHouse services in compliance with relevant export control and data privacy laws

- Maintain relevant independent third-party audits, standards, and certifications